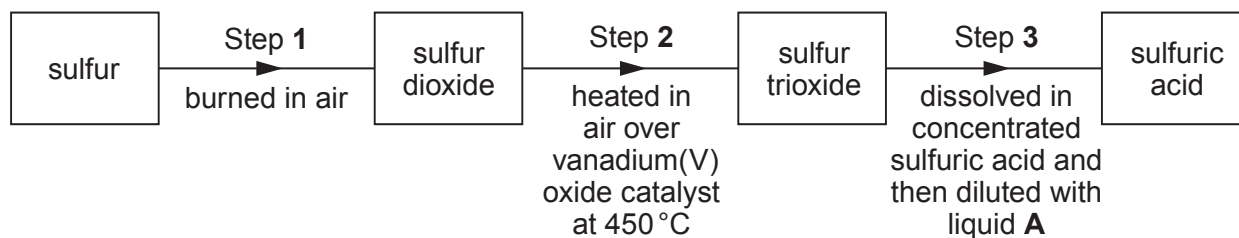
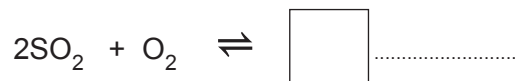


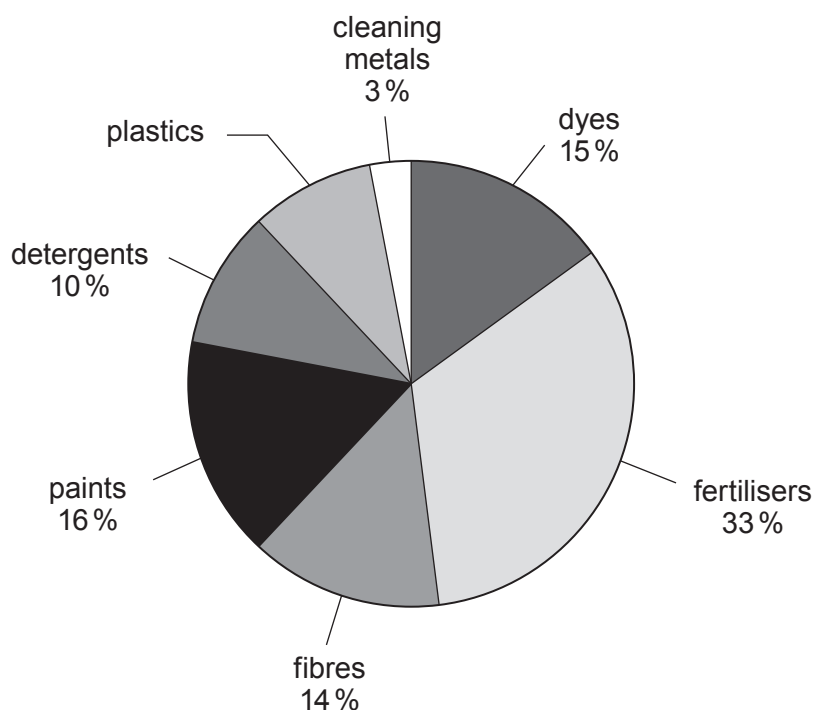
3. (a) The flow diagram shows steps in the manufacture of sulfuric acid using the Contact Process.



- (i) Name the **element** in the air which reacts with sulfur in step 1. [1]
.....
- (ii) State the purpose of the catalyst in step 2. [1]
.....
- (iii) Name liquid **A** used to dilute the concentrated sulfuric acid in step 3. [1]
.....
- (iv) Complete and balance the symbol equation for the reaction in step 2. [2]



(b) The pie chart shows the main uses of sulfuric acid.



Calculate the percentage of sulfuric acid used for making plastics.

[2]

Percentage = %

(c) One important use of sulfuric acid is in the production of the fertiliser ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.

Complete the following word equation for the production of ammonium sulfate.

[1]

sulfuric acid + $\longrightarrow$ ammonium sulfate

